

“Teaching is like brain surgery without breaking the skin. It should not be entered into lightly nor performed by amateurs”.

Neurologic Emergencies

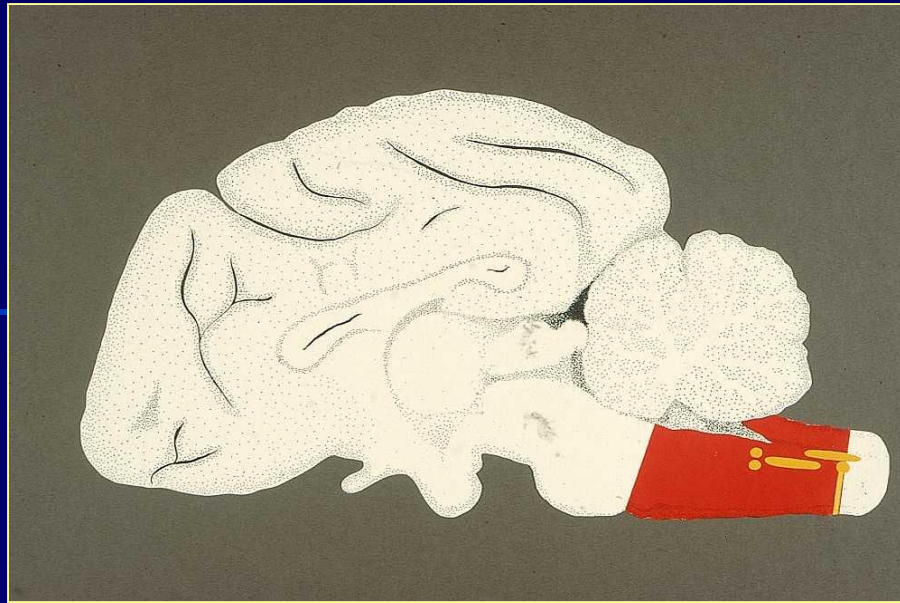
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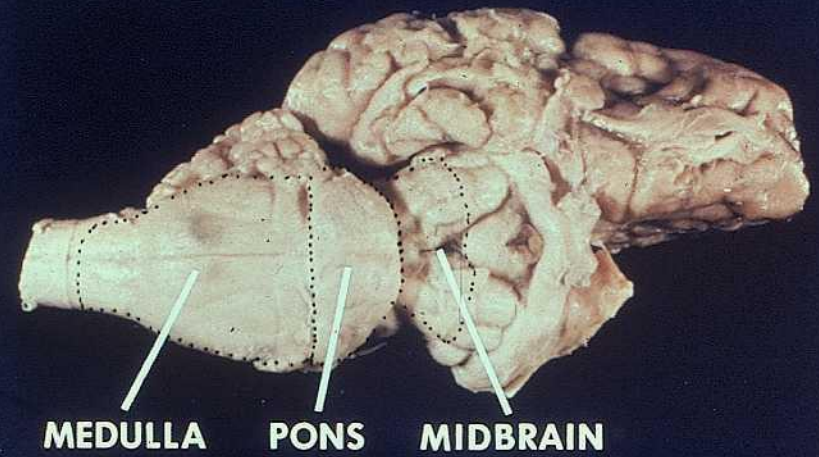
Neurologic Emergencies

What are they ?

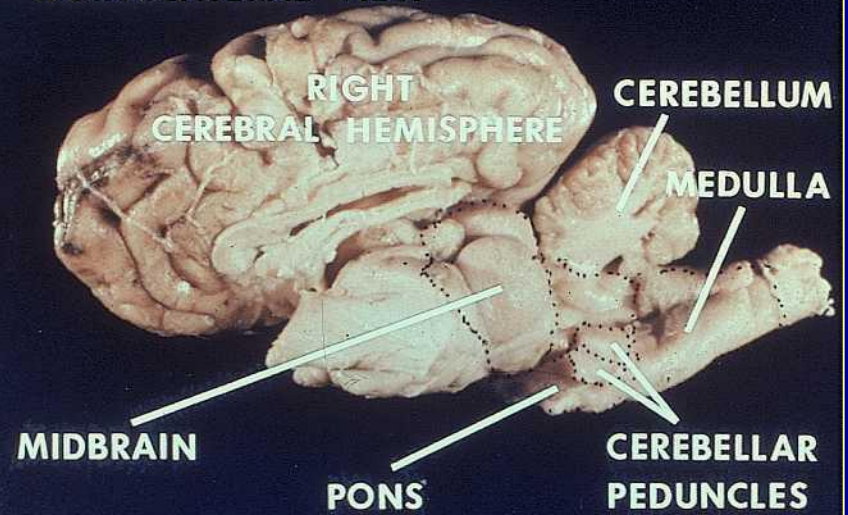
- Any structural or functional disturbance of the CNS which has an acute onset
 - Can result in reversible neuronal impairment
 - Must be treated aggressively to avoid irreversible changes or death



VENTRAL VIEW

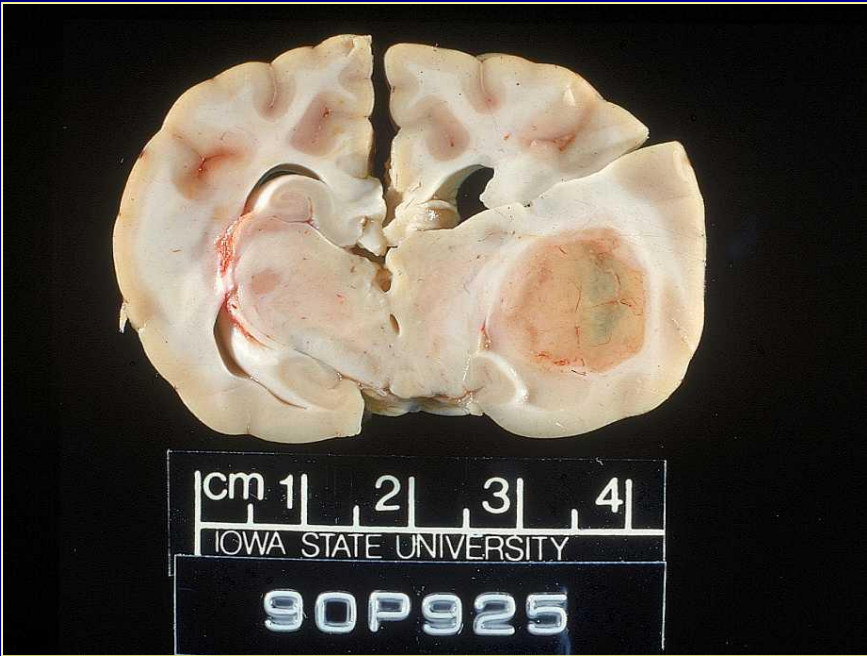


DORSOLATERAL VIEW



Types of Neurologic Emergencies

- ☛ Intracranial
 - Traumatic - HBC
 - Metabolic - Hypoglycemia / HE
 - Neoplastic
 - Vascular
 - Seizures
- ☛ Spinal Cord - IVDD, FCE, Trauma
- ☛ Neuromuscular - Myasthenia gravis

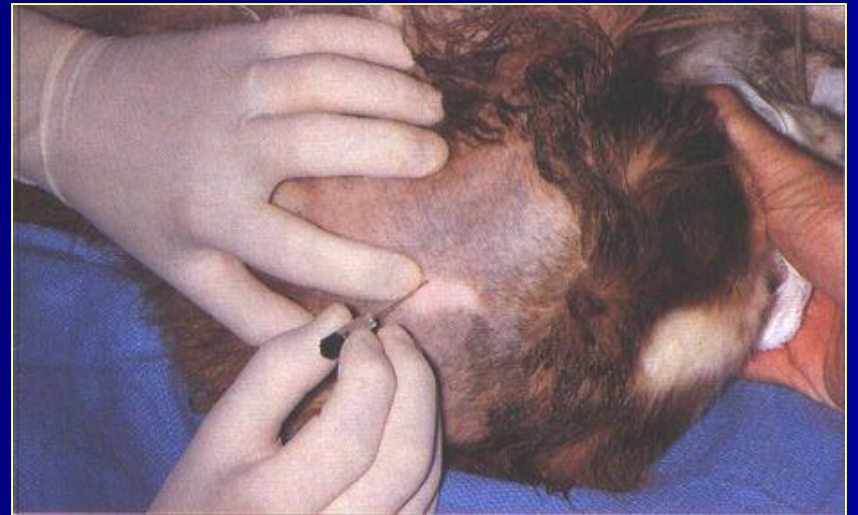


Neurologic Assessment

- ◀ Mentation
 - ◀ Level and content of consciousness
- ◀ Cranial nerves
- ◀ Gait
- ◀ Postural reactions
- ◀ Motor
- ◀ Sensory

Importance of the Neurologic Exam

- ◀ Neurolocalization
- ◀ Prognostication
- ◀ Very important to do serial neurologic examinations to look for progression



Traumatic Brain Injury

- ☛ HBC / Missile injuries / Falls / Bite wounds
- ☛ Progressive increase in intracranial pressure (ICP) most deadly
- ☛ Remarkable compensatory powers
- ☛ Patience is a virtue



Primary brain injury

- Physical disruption of intracranial structures that occurs immediately post-trauma
- Direct damage to brain parenchyma
 - Contusions / lacerations / diffuse axonal injury
- Intracranial hemorrhage / vasogenic edema

Primary brain injury

- Extent is a function of force of impact
- Acceleration / deceleration forces of impacting objects / intracranial contents
- Direct parenchymal damage ☹️

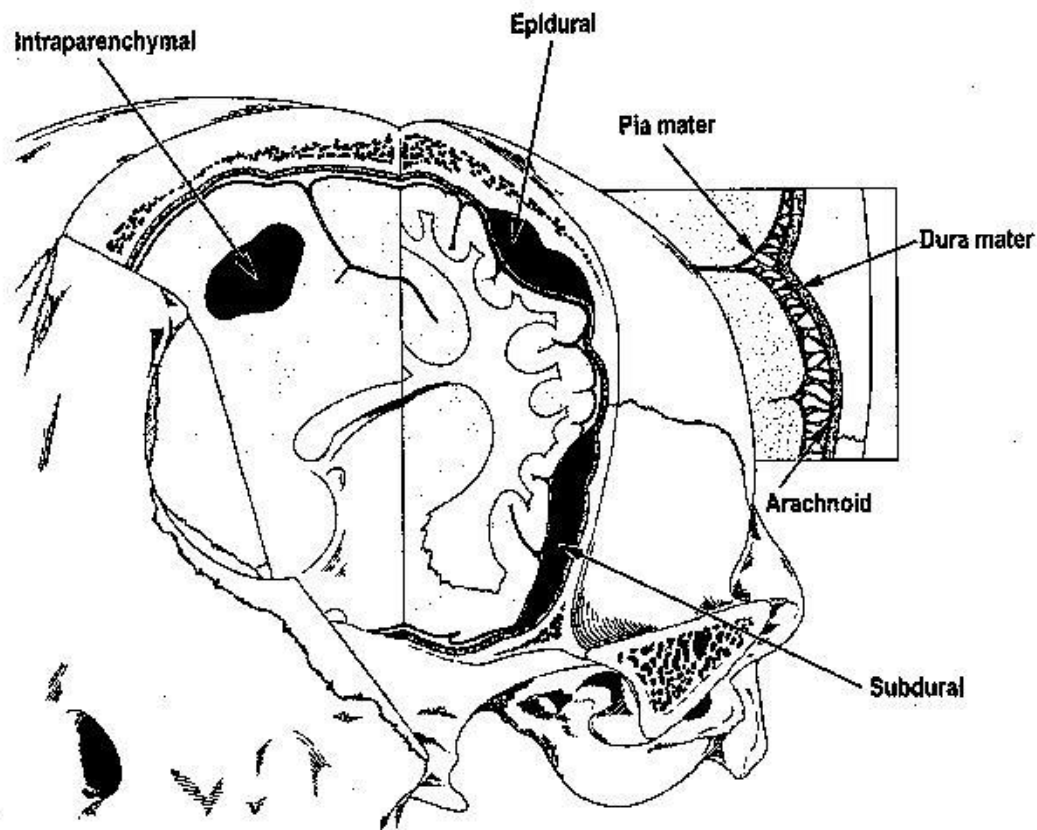
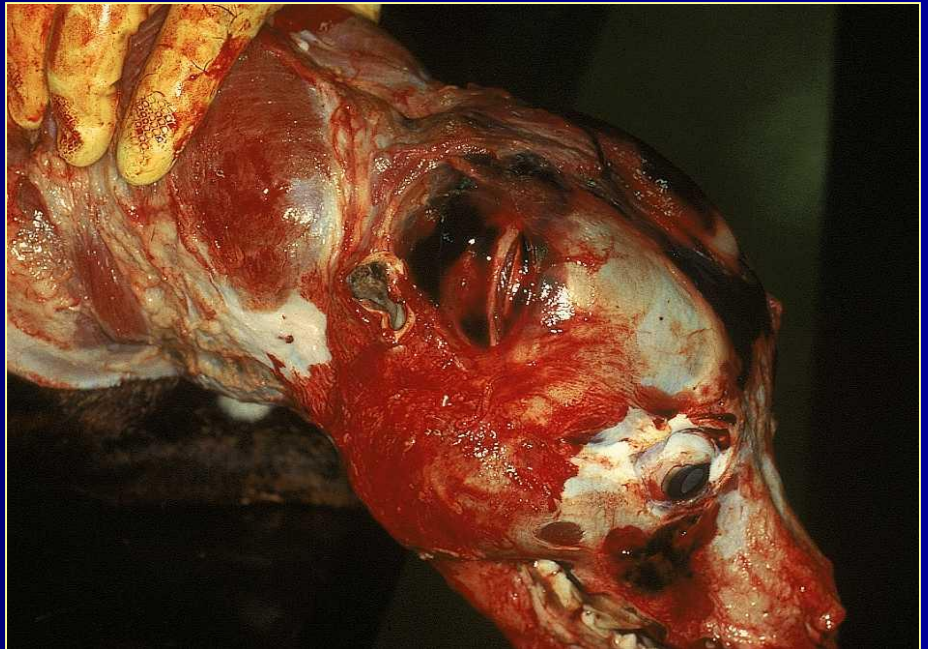
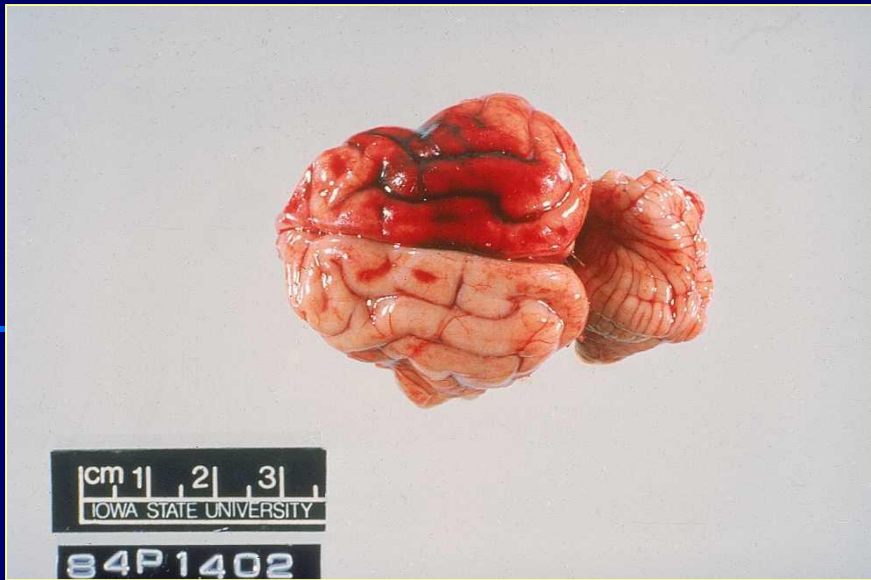


Fig. 5.1. Clinically important forms of intracranial hemorrhage. Subarachnoid hemorrhage (not depicted) would occur as diffuse hemorrhage between the pia and arachnoid layers (see inset)(Illustration by Anton Hoffman).



Secondary brain injury

- Continued hemorrhage and edema
- Chain of events lead to brain-tissue damage / increased ICP
- ATP depletion – disruption of cellular ionic homeostasis
- Na and Ca influx – leads to cellular swelling (cytotoxic edema) / depolarization

Secondary brain injury

- Uncontrolled depolarization >> glutamate released extracellularly
- Leads to continued increases in intracellular calcium levels
- Leads to activation of tissue-damaging pathways
 - Arachidonic acid cascade (phospholipase A2 activation)
 - Xanthine oxidase pathway (free-radical production)

Secondary brain injury

- Iron – vital cofactor in xanthine oxidase pathway >> free radical *spp*
 - Hydroxyl and superoxide radicals
- These radicals damage cell membranes high in polyunsaturated fatty acids (PUFA's) and cholesterol
 - Brain tissue high in Fe & membranes with high PUFA's and cholesterol

Secondary brain injury

- Free radicals
 - Very damaging to neuronal membranes
 - Production induced by:
 - Ischemia
 - Arachidonic acid metabolites
 - Catecholamine oxidation
 - Activated neutrophils

Secondary brain injury

- Other autolytic processes induce the following cascades:
 - Complement
 - Kinin
 - Coagulation / fibrinolytic
- Elevated NO (nitric oxide) / cytokines (TNF / interleukines)
- Self-perpetuating .. Can lead to herniation
- Lactic acid accumulation (anaerobic glycolysis)
- Above results in increased ICP >> clinician has some control over this ☺

Intracranial pressure (ICP) dynamics

- ICP = pressure exerted by tissues/fluids within the cranial vault
 - Normal ICP = 5-12 mm Hg
 - CPP (cerebral perfusion pressure) is a primary determinant of CBF (cerebral blood flow), thus brain oxygenation/nutritional support
 - $CPP = MABP - ICP$

Intracranial pressure dynamics

- Normal contents of cranial cavity:
 - Brain tissue (80%)
 - Blood (10%)
 - CSF (10%)
 - ICP remains constant if above in equilibrium
 - MABP 50-150 mm Hg – ICP remains constant – *Pressure autoregulation*
 - Links systemic BP changes to blood vasculature tone
 - If MABP rises >> brain vasoconstriction / if MABP falls >> brain vasodilation
 - In normal animal, the former prevents ICP from rising / the latter prevents ICP from falling

Intracranial pressure dynamics

- Chemical autoregulation
 - Direct responsiveness of brain to partial pressure of CO₂ in arterial blood
 - Increased PaCO₂ >> vasodilation
 - Decreased PaCO₂ >> vasoconstriction

Intracranial pressure dynamics

- **Monro-Kellie principle (intracranial compliance)**
 - Expansion of one component requires = decrease in other compartments to maintain normal ICP
 - Following head trauma >> IC hemorrhage / edema increase volume in the skull
 - Initially, CSF / blood flow altered to accommodate increased volume
 - IC compliance overwhelmed >> ICP increases exponentially (low IC compliance)

Intracranial pressure dynamics

- If hypotension (MABP) / hypoxia / hypercapnea develops >> brain vasculature dilates to preserve CBF
 - Serves to increase ICP, but CPP / CBP remain inadequate
- Increasing ICP >> further ischemia / edema >> herniation

Brain herniation

- Secondary effects accumulate
- Intracranial compliance (change in volume per unit change in pressure) decreases as ICP increases
- CPP compromised ... herniation (sound familiar ?)
- Cushings Reflex – low heart rate and high blood pressure – intracranial hypertension

Brain herniation

- Brain volume / ICP increase beyond limits of compensation
- Parenchyma shifts causing brain herniation

Brain herniation

- Caudal transtentorial
- Rostral transtentorial
- Foramen magnum
- Subfacial
- Through a fracture site

Brain herniation

- Caudal transtentorial
 - Temporal / occipital cortex protrudes under the tentorium cerebelli causing compression and caudal displacement of the midbrain
- Rostral transtentorial
 - Cerebellum displaced rostrally under the tentorium cerebelli compressing and displacing the temporal cortex and brainstem
- Watch for bilateral CN III compression

Brain herniation

- Foramen magnum herniation
 - Caudal cerebellar vermis is displaced through the foramen magnum resulting in malacia and hemorrhage to the cerebellum and medulla oblongata
 - Respiratory arrest >> respiratory centers in the medulla oblongata

Brain herniation

■ Subfacial

- Caudal cerebellar vermis is displaced under the falx cerebri
- The opposite cingulate gyrus is compressed with minimal neurologic dysfunction apparent

Other brain injury

- Coup injury refers to direct injury by external force at the site of impact
- Contrecoup injury is common when the head is in motion or after the initial coup
- Closed head injury refers to brain injury without any penetrating injury to the brain
- Open head injury – bone fragments and foreign material being driven into the brain which could lead to infection

Parameters that should be monitored in brain injury

- Blood pressure
- Body temperature
- HR / Rhythm
- RR / character
- Serum glucose / osmolality
- Electrolytes
- Lactate
- SpO₂ / PaO₂
- PaCO₂
- ICP – intracranial pressure monitor

Initial Assessment and Therapy

- ☛ ABC's of trauma management
- ☛ Quick assessment tests
- ☛ Immediate correction of hypoxemia and hypovolemia to avoid increased ICP
- ☛ Act calmly but quickly

Initial assessment

- Glasgow Coma Scale
 - Human GCS is graded from 3-15
 - Scores below 8 indicate a severe brain injury
 - CT scan allows accurate assessment of skull / brain injuries – epidural hematoma
- Modified GCS (MGCS)
 - Assesses level of consciousness / brainstem reflexes / motor activity on a scale of 3-18
 - Each of these 3 examination categories is assigned a score from 1-6
 - ARAS



Initial assessment

- Hypotension and hypoxia must be identified and avoided
- Blood glucose concentration – euglycemia should be maintained
- Hyperglycemia has been correlated with severity of head trauma in dogs and cats

Fluid Therapy

- ◀ Fluid therapy is indicated to support CPP by volume replacement / maintenance of MABP
- ◀ Limitation of fluid replacement is CONTRAINDICATED
- ◀ Hypotension has been repeatedly shown to be a reliable predictor of sustained elevations of ICP and increased mortality in human head trauma victims

Fluid therapy

- Maintain blood pressure at 90 mm Hg systolic pressure
- Avoid hypo / hypervolemia
- Volume replacement fluids
 - Hypertonic saline can improve MABP and thus CPP w/o exacerbating brain edema
 - Very useful in cases of traumatic brain injury

Fluid therapy

- Hypertonic saline (7 %): 4-5 ml/kg over 3-5 minutes for shock
- Be careful of sodium >> implicated as a major osmotic agent contributing to brain edema
- Focal areas of damaged parenchyma ?

Fluid therapy

- Dextran 70: 10-20 ml/kg (up to 40-50 ml/kg/hr) for shock in dogs
- Cats: 5 ml/kg boluses given over 5-10 minutes – max of 20 ml/kg
- Less than beneficial compared to above
- Potential adjunctive therapy choice

Fluid therapy

- Crystalloids (LRS, 0.9% saline): 90 ml/kg/hr (dogs), 60 ml/kg /hr (cats) for shock
- Watch for overhydration / brain edema / increased ICP
- Fluid administration should be tapered when physiologic goals met
- Blood product administration anemia
 - 4-10 ml/kg/hr (over 4-6 hr) in the stable patient

Oxygenation and hyperventilation

- Hyperoxygenation recommended for acute brain injury
- Maintain arterial blood gas at PaO₂ of or above 90 mm Hg for dogs (100 for cats)
- Pulse oximetry

Oxygenation and ventilation

- O2 supplementation
 - Face mask / nasal O2 / transtracheal / O2 cage
- Loss of consciousness and ventilation
- Maintain PCO2 levels between 25-35 mm Hg
- Avoid overventilation and vasoconstriction

Other therapies

- Head elevation: Maximize cerebral venous drainage >> 30 degrees
- Address cerebral edema >> peaks at 24-48 hours post-injury and can persist for almost 96 hours
- Physical therapy/ Nutritional support

Mannitol (20-25%)

- Osmotic diuretic
 - Decreases cerebral edema and ICP while maintaining CPP
 - Reduces cerebral edema through osmotic mechanisms
 - Decreases CSF production
 - ***Reflex vasoconstriction*** of brain vasculature by decreasing blood viscosity and free radical scavenging
 - Brain's pressure autoregulation mechanism (allows for improved CPP at a lower brain-blood volume – decreased ICP) – occurs in minutes

Mannitol

- Dose: 0.5 – 1.5 g/kg IV over 10-20 minutes
- Reaches maximal osmotic effect 10-20 minutes after administration with effects lasting 2-5 hours
- Limit to 3 boluses in a 24 hour period
- Do not use in the face of hypovolemia
- Can repeat every 6-8 hours / monitor serum osmolality / electrolytes

Other Therapeutics

- ☞ Lasix (Furosemide)
 - ☞ Loop diuretic
 - ☞ Dose prior to mannitol administration at 2-5 mg/kg IV to decrease rebound of ICP
- ☞ Glucocorticoids - use is questioned
 - ☞ "High-dose Methylprednisolone" protocol
 - ☞ 30 mg/kg at time 0, then 15 mg/kg at 2 and 4 hours IV
 - ☞ Thought to scavenge free radicals vs. activating steroid receptors
 - ☞ Brain trauma Foundation guidelines state that glucocorticoids are not recommended for improving outcome or reducing ICP in human head trauma patients

Other therapies

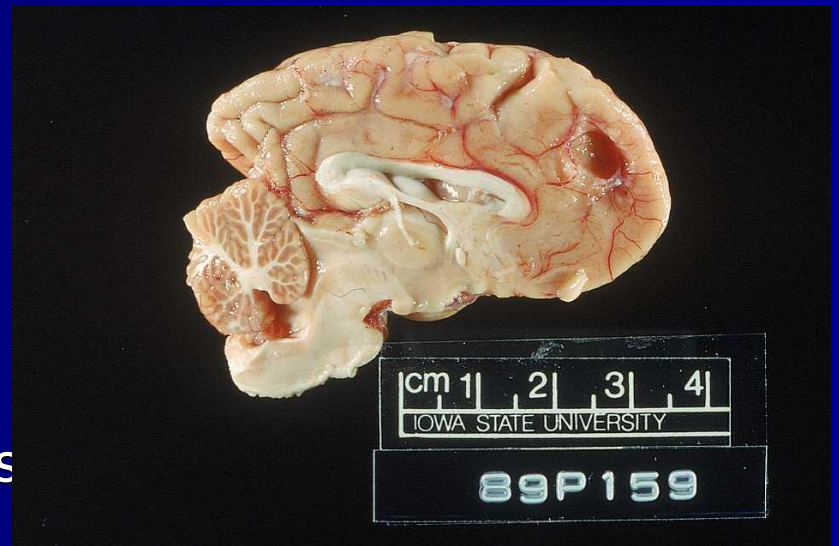
- **Lazaroids**
 - 21-aminosteroids
 - Methylprednisolone analogues
 - Do not activate glucocorticoid receptors
 - Benefits of free radical scavenging
- **Avoid hyperglycemia**
 - Fuels anaerobic metabolism >> brain lactic acid
 - Associated with increased mortality in humans
 - Recent veterinary study found it to be correlated with severity of neurologic dysfunction, but association with outcome not found

Other therapies

- Free-radical scavengers
 - Dimethylsulfoxide (DMSO)
 - Allopurinol
 - Deferoxamine mesylate
 - Liposome-encapsulated forms of superoxide dismutase/catalase
 - Lack of clinical evidence to support their use currently
- Erythropoietin
- Antagonists of opiate / glutamate receptors
- Hypothermia – human studies (89.6-93.2 degrees)
 - Thought to inhibit inflammatory cytokine/glutamate release

Brain Neoplasia

- Signs observed due to vasogenic edema
- Extremely steroid responsive
- Hypertonic saline 3 ml/kg IV over 20 minutes
- Mannitol
 - Dose: 0.5-1 g/kg IV slow over 10-20 minutes
- Dex SP 0.15 mg/kg IV once



Secondary assessment and diagnostic procedures

- Thorough neurologic examination once the patient is stabilized
- CT scan is the standard of care in human head trauma patients
 - Availability
 - Rapid scan times
 - Better visualization of peracute hemorrhage / bony abnormalities / fractures
- Specific guidelines for therapeutic intervention exist for open/depressed skull fractures

Diagnostic Imaging



Fig. 5.5. Noncontrast, computed tomographic (CT) brain image of a brain-injured dog. Note the evidence of intraparenchymal hemorrhage (Reprinted with permission!).



Fig. 5.4. Depressed skull fracture in a dog with deteriorating neurologic status following severe head trauma (Reprinted with permission¹²).

Evaluation of the Neurologic Patient

- Serial neurologic exams are a must !!
- Look for trends in consciousness, breathing patterns, pain perception and cranial nerve function
- Can indicate worsening of condition or progression to brain herniation
- Take nothing for granted !!

Neurologic Exam

- Loss of consciousness
- PLR
- Vision
- Vestibuloocular movements
- Motor function
- Respiratory pattern

Pupil size and Symmetry

- Normal direct and indirect response
- Unilateral oculomotor lesion
- Midbrain compression
- Bilateral oculomotor lesion
- Diffuse forebrain lesion



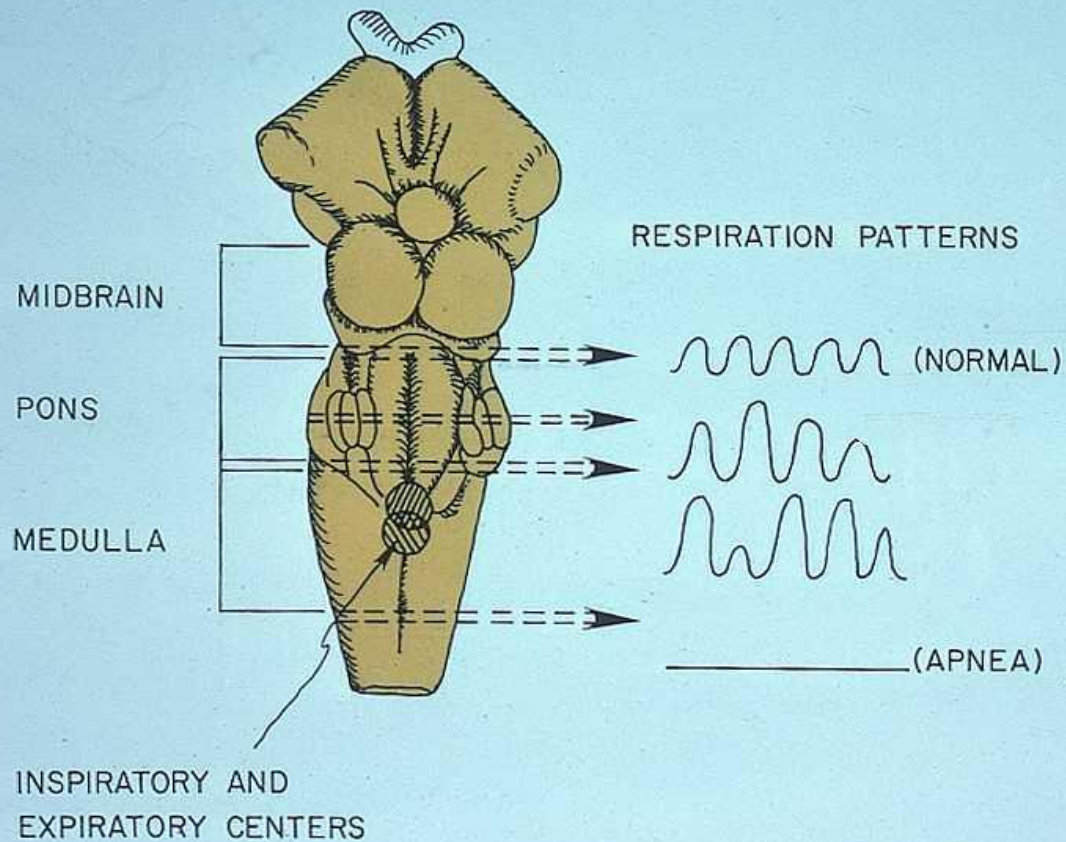
Vestibuloocular Movements

- Strabismus
- Presence or absence of ocular movements
- Presence of nystagmus
 - Sustained or resting
 - Positional or changing

Respiratory Patterns

- Cheyne-Stokes
 - Implies bilateral dysfunction
 - Deep cerebral/diencephalon
- Central neurogenic hyperventilation
 - Mesencephalic lesion
- Irregular / Ataxic breathing
 - Caudal brainstem/medullary lesion

BRAIN STEM CONTROL OF RESPIRATION



Motor Function

- Upper Motor Neuron reflexes
 - Fore > Rear : Deep cerebral
 - Extensor rigid fore > rear : Midbrain
 - Extensor rigid fore/opisthotonus:
Cerebellum
 - Fore/Rear : Pons/rostral medulla
 - Absent : Caudal medulla

Post-Traumatic Syndrome

- Hemispheric damage
- Brainstem injury / compression
- Cerebellovestibular dysfunction
- Epilepsy
- Diabetes insipidus
- Effects of :
hypovolemia/hypoxia/hypotension

Surgical intervention - Humans

- Indications for intervention are well defined in human head trauma treatment
- Include presence and extent of focal intracranial hemorrhage
- Size, location of focal hemorrhage / presence of a midline shift combined with ICP measurement and GCS – factors affecting surgical decisions in humans

Surgical intervention – veterinary medicine

- Indications for surgical intervention
 - Open or depressed skull fractures
 - Ongoing skull / scalp hemorrhage
 - Intracranial hemorrhage causing a mass effect
- Recent retrospective veterinary study describing CT findings in head trauma highlights the limitations of CT compared to MRI, necropsy and surgical findings
- Study did not identify any characteristic CT findings that correlated with survival
- Authors cautioned that CT findings should not be the only determinant in prognosis

Surgical intervention

- Value of craniotomy solely as a decompressive surgery is unknown in dogs and cats
- In normal dogs and cats, combined craniotomy / durotomy results in dramatic decreases in ICP
- Surgery should be considered if deterioration in neurologic status despite aggressive medical therapy
- ICP pressure monitoring
 - Humans – ICP below 20 mm Hg Cpp at a minimum of 70 mm Hg
 - Fiberoptic epidural ICP monitoring >>> future directions

Spinal Cord Emergencies

- ☞ Luxations / Subluxations
- ☞ Intervertebral Disc rupture
- ☞ Fibrocartilagenous embolism (FCE)
- ☞ Meningitis
- ◆ Remember that aortic thromboembolism can present with similar signs



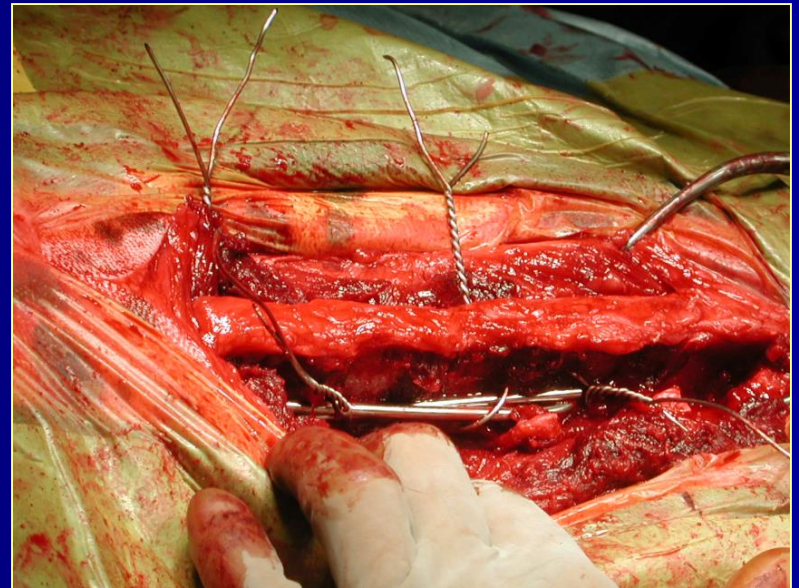
Spinal Cord Emergencies

- The value of neurolocalization
- C1 - C5
- C6 - T2
- T3 - L3
- L4 - S2
- Prognostication
- Keep patient as quiet as possible

Spinal Cord Emergencies

- Type of injury
- Duration of clinical signs
- Presence or absence of voluntary motor activity
- Presence or absence of conscious pain perception





Spinal Cord Emergencies

- If conscious pain perception is lost (both superficial and deep pain) for > 6 hours, prognosis drops considerably
- If present (depending on cause of injury), > 75-80 % chance of regaining function
- Conscious proprioception > voluntary motor > superficial pain > deep pain

Treatment for Spinal cord Emergencies

- Stabilize the patient - The ABC's of critical care
- Neurologic Assessment
- **Corticosteroids** – Use is controversial
 - Not advised
- **Polyethylene glycol** – membrane repair
- **Oscillating Field Stimulation** – Application of weak electrical field of oscillating polarity to damaged SC region – neuronal / axonal repair

Spinal Cord Injury

- Surgical intervention is indicated where appropriate
- Immediate referral is indicated when animal has lost voluntary motor or is deteriorating
- Beware of mistaking the withdrawal reflex for conscious pain perception

Key Points to Remember

- Complete history
- Thorough physical and neurologic exams
- ABC's of critical care
- Do not be too hasty on prognostication - look for trends
- Treat the animal, not its blood work
- Patience is a virtue



There are some simple truths ... and the dogs know what they are.

Joseph Duemer (b.1951)

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